香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

ECO6102 / 2026–27 Term 1 (Fall)

Lecture 4a: Instrumental Variables

Enjie (Jack) Ma

Lecture 4 · Mondays 12 and 19 October 2026

Instrumental variables across two lectures

Our teaching text for IV is Wooldridge, Introductory Econometrics, sixth edition

We use *Introductory Econometrics*, Chapter 15, for these two lectures. The graduate book's Chapter 5 supplies optional formal detail.

Lecture 4a: Sections 15-1–15-2

Motivating example; one instrument: assumptions, estimation, uncertainty and controls.

Lecture 4b: Sections 15-3–15-6

2SLS with several instruments, weak instruments and specification tests.

Our emphasis is intuition and the interpretation of assumptions.

Today's contents

Intro 15-1–15-2

1. Motivation: Omitted Variables in a Simple Regression Model
2. Statistical Inference with the IV Estimator
3. Properties of IV with a Poor Instrumental Variable
4. Computing R-Squared after IV Estimation
5. IV Estimation of the Multiple Regression Model

Some Terminology

Keep the population quantity separate from the sample calculation

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

Parameter / estimand

β_1: the population schooling effect we want to learn.

Estimator

$\hat{\beta}_1$: a rule that turns sample data into an answer; it varies across samples.

Estimate

A realized number, such as .059, obtained from one sample.

Population ↔ Sample

The population target stays fixed while the estimate changes

Random sample	**Estimator: OLS rule**	**Estimate**
$\{(x_i, y_i)\}_{i=1}^{n}$	$\hat{\beta}_{1,OLS} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$	One realized number
Schooling and log wage		$\hat{\beta}_{1,OLS} = .109$

Population target: β_1. A new sample can give a different estimate.

Statistical properties describe the estimator across possible samples.

Contents

Intro 15-1

1. Motivation: Omitted Variables in a Simple Regression Model

Instrument conditions and candidate instruments

Identification and the IV estimator

2. Statistical Inference with the IV Estimator

3. Properties of IV with a Poor Instrumental Variable

4. Computing R-Squared after IV Estimation

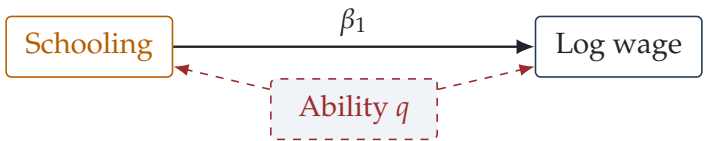
5. IV Estimation of the Multiple Regression Model

Revisiting OVB

Running wage example: OLS can mix schooling with ability

$$\log(wage) = \beta_0 + \beta_1 educ + \underbrace{\gamma q + v}_{u}, \qquad q = ability$$

Blue: instrument candidate Orange: endogenous Red: omitted



Two approaches to omitted ability

Proxy controls for ability; IV changes the variation used

Same target
$\log(wage) = \beta_0 + \beta_1 educ + \gamma q + v$.

1. Proxy Add a measured proxy p for q. It must capture the ability variation that is related to schooling.

If a suitable proxy is unavailable, keep ability in the error and look for a valid instrument.

2. Instrumental variable Use schooling variation linked to an instrument z with $\text{Cov}(z, u) = 0$.

Recall the proxy conditions

Contents

Intro 15-1: instrument conditions

1. Motivation: Omitted Variables in a Simple Regression Model

Instrument conditions and candidate instruments

Identification and the IV estimator

2. Statistical Inference with the IV Estimator

3. Properties of IV with a Poor Instrumental Variable

4. Computing R-Squared after IV Estimation

5. IV Estimation of the Multiple Regression Model

Two conditions for an instrument z

Assumption 1: instrument exogeneity; assumption 2: relevance

$$y = \beta_0 + \beta_1 x + u, \qquad E(u) = 0$$

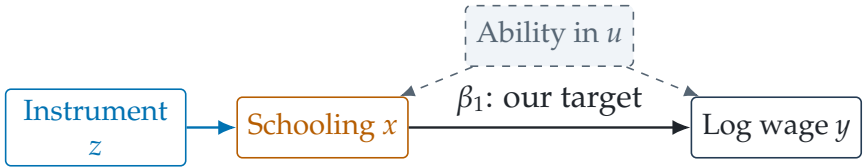
1. Instrument exogeneity: $\text{Cov}(z, u) = 0$ (15.4)
2. Instrument relevance: $\text{Cov}(z, x) \neq 0$ (15.5)

IV changes the source of variation

The schooling effect is the target; omitted ability still exists

Wage equation

$\log(wage) = \beta_0 + \beta_1 educ + u.$



Ideal comparison Change schooling; hold u fixed.

Observed data Schooling can move with omitted ability.

IV Use z-linked schooling variation, with $\text{Cov}(z, u) = 0$. This supports consistency; ability still matters.

Verifying relevance + exclusion condition

Check the first stage; defend the remaining wage paths

$$x = \pi_0 + \pi_1 z + r, \qquad \pi_1 = \frac{\text{Cov}(z, x)}{\text{Var}(z)}$$

- **Relevance** Regress x on z; examine the estimated sign, magnitude and uncertainty.
- Rejecting $H_0 : \pi_1 = 0$ is a minimum check; it need not imply a strong instrument.
- **Exogeneity** $\text{Cov}(z, u) = 0$. Because u is unobserved, we generally cannot test this directly.
  - Defend no separate effect of z on the outcome outside x.
  - Defend no association between z and omitted causes in u, using economic behavior and the assignment process.

Relevance in the schooling example

Does the instrument predict differences in schooling?

Target
$\log(wage) = \beta_0 + \beta_1 educ + u$, with ability in u.

Candidate instrument Father's schooling, *fatheduc*.

1. $\text{Cov}(fatheduc, u) = 0$,
2. $\text{Cov}(fatheduc, educ) \neq 0$.

Fathers with more schooling may encourage or support more schooling for their children.

$$educ = \pi_0 + \pi_1 fatheduc + v, \qquad H_0 : \pi_1 = 0$$

Check the expected positive relationship; assess exogeneity separately.

Exogeneity in the schooling example

Exclude a separate wage effect of the instrument

Target

$\log(wage) = \beta_0 + \beta_1 educ + u.$

$$\text{Cov}(fatheduc, u) = 0$$

Exclusion Father's schooling must have no separate effect on wages once own schooling and omitted wage determinants are held fixed.

For example, family connections could raise wages through a channel other than own schooling. That would threaten exclusion.

Exogeneity also concerns omitted causes - ability

A schooling mechanism can support relevance and threaten validity

Target

$\log(wage) = \beta_0 + \beta_1 educ + u$, with ability in u.

$$\text{Cov}(fatheduc, u) = 0$$

Father's schooling must also be unrelated to ability and other omitted wage determinants. Family background makes this assumption debatable.

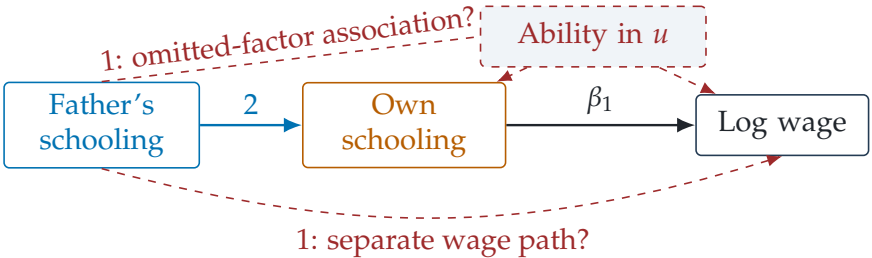
We can observe the schooling relationship. We need a separate argument about these unobserved wage determinants.

Locate the two instrument conditions

Relevance uses the schooling link; exogeneity rules out other wage paths

Maintained conditions

1. Cov(*fatheduc*, *u*) = 0 2. Cov(*fatheduc*, *educ*) ≠ 0.



Defend condition 1 by ruling out the two red instrument links.

Choosing a IV candidate - Not Easy!

One condition can hold while the other fails

Wage equation

$\log(wage) = \beta_0 + \beta_1 educ + u.$

Which condition is at risk for each candidate?

Candidate	Relevance $\text{Cov}(z, educ) \neq 0$	Exogeneity $\text{Cov}(z, u) = 0$
SSN last digit	No schooling mechanism	Plausibly unrelated to ability
IQ	Related to schooling	Tracks ability in u

A good ability proxy is usually a poor schooling IV: it tracks ability.

Family background as an instrument

Plausible schooling links still need an exogeneity argument

Wage equation

$\log(wage) = \beta_0 + \beta_1 educ + u.$

Does a plausible schooling link settle instrument validity?

Candidate	Relevance	Exogeneity concern
Mother's schooling	Usually positive	May track ability and nurturing
Number of siblings	Usually negative	Other family wage paths

Skipping classes and exam scores

Distance may predict absence, but family income can threaten exogeneity

$$score = \beta_0 + \beta_1 skipped + u \tag{15.8}$$

Concern Ability and motivation may affect both attendance and scores.

Candidate instrument

Distance from living quarters to campus may increase missed classes. Check the expected positive first-stage relationship.

Exogeneity concern

Family income may affect both where a student lives and exam performance. Distance could then be correlated with u.

Section 15-2 will show how to include observed controls.

Contents

Intro 15-1: identification and estimation

1. Motivation: Omitted Variables in a Simple Regression Model

Instrument conditions and candidate instruments

Identification and the IV estimator

2. Statistical Inference with the IV Estimator

3. Properties of IV with a Poor Instrumental Variable

4. Computing R-Squared after IV Estimation

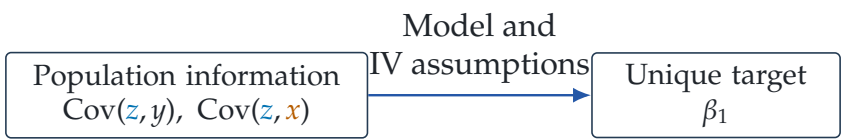
5. IV Estimation of the Multiple Regression Model

Recall identification

Would the population information determine one value of the target?

$$y = \beta_0 + \beta_1 x + u$$

Identification Under the model and maintained assumptions, the population distribution of the observable variables (y, x, z) determines one value of β_1.



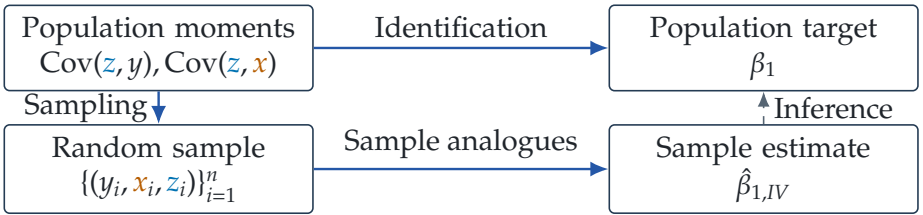
Identification is a population question. We still need a sample to estimate those moments.

From identification to an estimate

First find the population answer, then use a sample to estimate it

Target

$y = \beta_0 + \beta_1 x + u$. We want the population slope β_1.



Inference Use sampling variation to assess uncertainty about β_1.

Identification with IV

Population moments determine the target under the assumptions

Identification We can write β_1 in terms of population moments that can be estimated using a random sample from the population.

$$y = \beta_0 + \beta_1 x + u, \qquad \text{Cov}(z, u) = 0$$

$$\text{Cov}(z, y) = \text{Cov}(z, \beta_0 + \beta_1 x + u)$$

$$\text{Cov}(z, y) = \beta_1 \, \text{Cov}(z, x) + \underbrace{\text{Cov}(z, u)}_{=0 \text{ by exogeneity}}$$

The identified population slope

Relevance makes the covariance ratio well defined

Model and conditions

$y = \beta_0 + \beta_1 x + u$.

$\text{Cov}(z, u) = 0, \quad \text{Cov}(z, x) \neq 0.$

$$\beta_1 = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)} \tag{15.9}$$

- **Exogeneity** Removes the error covariance.
- **Relevance** Makes division possible.
- Both remaining moments involve observable variables.

Next, replace these population moments with sample analog.

A covariance is a population moment

The expectation tells us which sample average to calculate

Identified target

$\beta_1 = \text{Cov}(z, y) / \text{Cov}(z, x)$.

Numerator

$$\text{Cov}(z, y) = E[(z - Ez)(y - Ey)]$$

$$\Downarrow$$

$$\frac{1}{n} \sum_{i=1}^{n} (z_i - \bar{z})(y_i - \bar{y})$$

Denominator

$$\text{Cov}(z, x) = E[(z - Ez)(x - Ex)]$$

$$\Downarrow$$

$$\frac{1}{n} \sum_{i=1}^{n} (z_i - \bar{z})(x_i - \bar{x})$$

Population averages become sample averages; population means become sample means.

The IV estimator

Replace population covariances with sample covariances

$$\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^{n}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{n}(z_i - \bar{z})(x_i - \bar{x})} \tag{15.10}$$

$$\hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV}\bar{x}$$

- The intercept makes the fitted line pass through $(\bar{x}, \bar{y})$.
- With exogenous x, using $z = x$ gives the OLS estimator.

Taking stock: IV assumptions and estimator

Identify the population slope, then estimate it from a sample

Model and conditions

$y = \beta_0 + \beta_1 x + u, \quad \text{Cov}(z, u) = 0, \quad \text{Cov}(z, x) \neq 0.$

Identified target $\beta_1 = \text{Cov}(z, y) / \text{Cov}(z, x)$.

$$\hat{\beta}_{1,IV} = \frac{\sum_i (z_i - \bar{z})(y_i - \bar{y})}{\sum_i (z_i - \bar{z})(x_i - \bar{x})} \xrightarrow{p} \beta_1$$

With random sampling and finite moments, IV is consistent.

Next, quantify sampling **uncertainty** with a standard error.

Why the sample ratio is consistent

Contents

Intro 15-1a

1. Motivation: Omitted Variables in a Simple Regression Model

2. Statistical Inference with the IV Estimator

Precision and the return to education

Binary instruments and binary treatments

3. Properties of IV with a Poor Instrumental Variable

4. Computing R-Squared after IV Estimation

5. IV Estimation of the Multiple Regression Model

Homoskedasticity for IV inference

The condition is stated relative to the instrument

$$y = \beta_0 + \beta_1 x + u, \quad E(u) = 0, \quad \text{Cov}(z, u) = 0, \quad \text{Cov}(z, x) \neq 0$$

$$E(u^2 \mid z) = \sigma^2 = \text{Var}(u) \tag{15.11}$$

This additional assumption concerns the structural error u and the instrument z.

With random sampling and the required finite moments, IV has a **large-sample normal approximation**.

Why the second moment equals the variance

What determines IV precision?

Noise raises uncertainty; variation and relevance reduce it

$y = \beta_0 + \beta_1 x + u$; valid z, homoskedastic errors.

$$\text{Avar}(\hat{\beta}_{1,IV}) = \frac{\sigma^2}{n \, \text{Var}(x) \, \rho_{xz}^2} \tag{15.12}$$

$\rho_{xz} = \text{Corr}(x, z)$.

Read the three ingredients of IV precision

Hold the other ingredients fixed when changing one

IV variance

$\text{Avar}(\hat{\beta}_{1,IV}) = \sigma^2/[n\,\text{Var}(x)\,\rho_{xz}^2]$.

- **Noise** σ^2
  - More structural noise ⇒ larger SE.
- **Spread** $\text{Var}(x)$
  - More regressor variation ⇒ smaller SE.
- **Relevance** ρ_{xz}^2
  - Stronger instrument–regressor association ⇒ smaller SE.

Estimating the ingredients of IV variance

Start with the estimated variance, then read its ingredients

$$\widehat{\text{Avar}}(\hat{\beta}_{1,IV}) = \frac{\hat{\sigma}^2}{SST_x \, R^2_{x,z}} \tag{15.13}$$

- $SST_x = \sum_i (x_i - \bar{x})^2$: SST_x/n estimates $\text{Var}(x)$.
- First-stage $R^2_{x,z}$ estimates ρ^2_{xz}.
- $\hat{\sigma}^2 = \sum_i \hat{u}_i^2/(n-2)$ estimates structural noise. We next calculate its residuals.

Structural residuals | R-squared and correlation

Use the structural equation for IV residuals

The coefficients come from IV; the regressor is the observed x

Structural model

$y_i = \beta_0 + \beta_1 x_i + u_i$.

$$\hat{u}_i = y_i - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV} x_i$$

Use **actual** x_i here. The instrument enters through the IV coefficient estimates.

Estimate noise from the structural residuals

Adjust for the two estimated structural coefficients

Structural residual

$\hat{u}_i = y_i - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV} x_i$.

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{u}_i{}^2$$

There are two estimated coefficients: the intercept and slope. This adjustment does not make the estimate exactly unbiased.

The IV standard error

Estimate the variance, then take its square root

Estimated variance

$\widehat{\text{Avar}}(\hat{\beta}_{1,IV}) = \hat{\sigma}^2/[SST_x\, R^2_{x,z}]$.

$$\text{se}(\hat{\beta}_{1,IV}) = \sqrt{\frac{\hat{\sigma}^2}{SST_x\, R^2_{x,z}}}$$

A smaller first-stage R^2 means a larger standard error, holding other ingredients fixed. This is the uncertainty we use in tests and intervals.

With SE → Hypothesis test and CI

The usual normal approximation is a large-sample claim

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\text{se}(\hat{\beta}_1)} \overset{a}{\sim} N(0,1) \quad \text{under } H_0 : \beta_1 = \beta_{1,0}$$

Approximate 95% CI: $\hat{\beta}_1 \pm 1.96\,\text{se}(\hat{\beta}_1)$

Example 15.1: the return to schooling

Schooling and omitted ability: same target, one common sample

$$\log(wage) = \beta_0 + \beta_1 educ + u$$

- **Sample** 428 working married women; hourly wage and years of education.
- **Instrument** Father's years of education, *fatheduc*.
- **Maintained claim** Father's education is uncorrelated with omitted determinants in u; family background makes this debatable.

Example 15.1: OLS benchmark

An extra schooling year is associated with about eleven percent more pay

Outcome model

$\log(wage) = \beta_0 + \beta_1 educ + u$. Same 428 working women.

OLS term	Coefficient	SE	t for zero
Intercept	−.1852	.1852	−1.00
Schooling	.1086	.0144	7.55

One more schooling year is associated with roughly 10.9% higher pay.

To claim 10.9% as causal requires schooling to be exogenous in this equation (clearly, its not!)

Example 15.1: first-stage evidence

Father's schooling predicts the daughter's schooling

First stage and relevance

$educ = \pi_0 + \pi_1 fatheduc + r$; $\text{Cov}(fatheduc, educ) \neq 0$. Same $n = 428$.

First-stage term	Coefficient	SE	t for zero
Father's schooling	.2694	.0286	9.43

An extra year of father's schooling predicts .27 more years for the daughter.

$\pi_1 \neq 0 \iff \text{Cov}(fatheduc, educ) \neq 0$.

✓ Evidence of relevance; exogeneity still needs an argument.

Why the slope measures relevance

Example 15.1: IV estimate

The estimated return is smaller and much less precise

Outcome model

$\log(wage) = \beta_0 + \beta_1 educ + u$; instrument: *fatheduc*, $n = 428$.

$$\hat{\beta}_{1,IV} = \frac{\widehat{\text{Cov}}(fatheduc, \log(wage))}{\widehat{\text{Cov}}(fatheduc, educ)} = .0592$$

IV term	Coefficient	SE	t for zero
Intercept	.4411	.4461	.99
Schooling	.0592	.0351	1.68

The estimated return is about 5.9%.

Example 15.1: an interval for the return

The coefficient's uncertainty matters for interpretation

Same wage equation

$\log(wage) = \beta_0 + \beta_1 educ + u$; IV estimate .0592, SE .0351.

$$95\%\ \text{CI:} \quad .0592 \pm 1.96(.0351) \approx [-.0096, .1280]$$

This is a large-sample interval under the maintained instrument and inference assumptions.

Example 15.1: what did IV change?

First compare the estimated schooling returns

Target

$\log(wage) = \beta_0 + \beta_1 educ + u$. Same 428 women.

Method	**Schooling slope**	SE
OLS	.1086	.0144
IV: father's education	.0592	.0351

$\hat{\beta}_{1,IV} = .0592 < .1086 = \hat{\beta}_{1,OLS}$: IV gives the smaller estimated return.

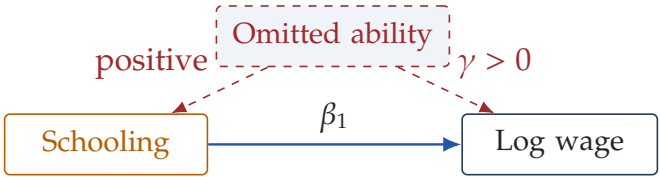
Can omitted-variable-bias logic explain why the OLS estimate is so much larger than IV?

Why omitted ability can push OLS upward

Schooling can receive credit for an omitted wage advantage

Model

$\log(wage) = \beta_0 + \beta_1 educ + \gamma\, ability + v.$



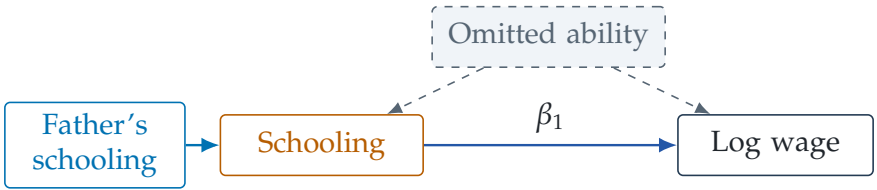
If ability raises both schooling and wages, OLS can give schooling credit for an omitted wage advantage.

IV focuses on variation linked to the instrument

The ability paths remain while our attention shifts

Maintained condition

$\log(wage) = \beta_0 + \beta_1 educ + u$; $\text{Cov}(fatheduc, u) = 0$.



IV uses the schooling variation linked to father's schooling. The ability paths still exist.

The omitted-ability bias in the equation

The positive covariance adds to the schooling slope

Model

$\log(wage) = \beta_0 + \beta_1 educ + \gamma\, ability + v.$

Assume $\text{Cov}(educ, v) = 0$.

$$\text{plim}\,\hat{\beta}_{1,OLS} = \beta_1 + \underbrace{\gamma}_{>0} \underbrace{\frac{\text{Cov}(educ, ability)}{\text{Var}(educ)}}_{>0} > \beta_1$$

Why both omitted-ability signs are positive

The two wage-example mechanisms explain the bias direction

Bias term

$\log(wage) = \beta_0 + \beta_1 educ + \gamma\, ability + v$.
OLS distortion: $\gamma \,\mathrm{Cov}(educ, ability)/\,\mathrm{Var}(educ)$.

- $\gamma > 0$: higher ability raises wages, holding schooling fixed.
- $\mathrm{Cov}(educ, ability) > 0$: people with higher ability tend to obtain more schooling.

IV estimate lower than OLS is compatible with this upward-bias story.

Example 15.1: compare the uncertainty

First observe the larger IV standard error

Target

$\log(wage) = \beta_0 + \beta_1 educ + u$. Same 428 women.

Method	Schooling slope	**SE**
OLS	.1086	.0144
IV: father's education	.0592	.0351

The IV standard error is about 2.44 times the OLS standard error in this sample.

Why is the IV estimate less precise?

Why IV can have a larger standard error

An exogenous-regressor benchmark: both OLS and IV are valid

Valid-OLS benchmark

$y = \beta_0 + \beta_1 x + u$, with $\text{Cov}(x, u) = 0$. Both OLS and IV are valid; assume homoskedasticity, random sampling and fixed nonzero relevance.

OLS

$$\text{Avar}(\hat{\beta}_{1,OLS}) = \frac{\sigma^2}{n\,\sigma_x^2}$$

Uses all regressor variation: x instruments itself.

IV

$$\text{Avar}(\hat{\beta}_{1,IV}) = \frac{\sigma^2}{n\,\sigma_x^2\,\rho_{xz}^2}$$

Uses the variation associated with z.

The precision cost of instrument-linked variation

A smaller instrument-regressor correlation raises the SE

Same valid-OLS benchmark

$y = \beta_0 + \beta_1 x + u$, $\text{Cov}(x, u) = 0$.

$$\frac{\text{Avar}(\hat{\beta}_{1,IV})}{\text{Avar}(\hat{\beta}_{1,OLS})} = \frac{1}{\rho_{xz}^2}, \qquad 0 < \rho_{xz}^2 \leq 1$$

IV's asymptotic SE is $1/|\rho_{xz}|$ times as large. A correlation of .5 doubles it.

Less instrument-linked variation means a larger SE.

A guide to an IV analysis

Use the wage example to organize the design before estimation

Start here

$\log(wage) = \beta_0 + \beta_1 educ + u$. Target: the return to one more schooling year.

1. **Name the concern** Ability in u may be related to schooling.
2. **Propose an instrument** Why might father's schooling predict own schooling?
3. **Defend exogeneity** What other wage paths or shared omitted causes could make this instrument invalid?
4. **Check relevance** Regress schooling on father's schooling. Examine sign, magnitude and uncertainty.

A guide to reading the IV results

Interpret the effect and its uncertainty under the stated assumptions

Same target and sample

$\log(wage) = \beta_0 + \beta_1 educ + u$. Compare OLS and IV for the same 428 women.

5. **Estimate** Report the IV coefficient and standard error. Keep OLS as a benchmark.
6. **Interpret** IV gives about a 5.9% return per schooling year, if the instrument assumptions hold.
7. **Compare carefully** Interpret the IV-OLS difference within the OVB angle.

After the break, we ask whether IV must always give a lower return.

Example 15.2: returns to education for men

An example where IV > OLS

WAGE2: 935 men, instrument: siblings

$\log(wage) = \beta_0 + \beta_1 educ + u.$

First stage

$$\widehat{educ} = 14.14 - \boxed{.228}\, sibs, \qquad SE(\hat{\pi}_1) = .030$$

One more sibling predicts about $\boxed{.23}$ fewer schooling years. (makes sense)

Relevance permits either sign: $\text{Cov}(sibs, educ) < 0$ is nonzero.

Example 15.2: compare the returns

IV can give a larger schooling coefficient than OLS

Same 935 men

$\log(wage) = \beta_0 + \beta_1 educ + u$. Instrument: number of siblings.

Method	Schooling slope	SE
OLS	.059	.006
IV: siblings	.122	.026

$\hat{\beta}_{1,IV} = .122 > .059 = \hat{\beta}_{1,OLS}$: IV gives a larger estimated return.

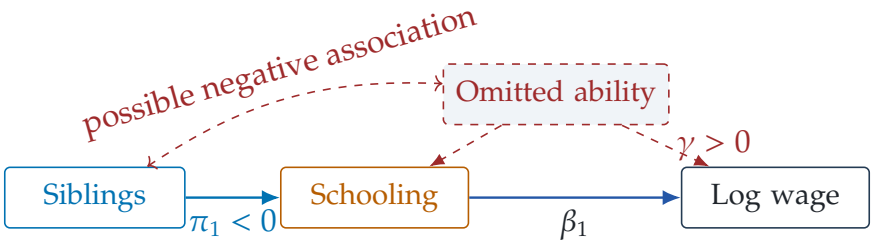
IV also has the larger SE. This differs from the coefficient ordering in the married women's sample.

Example 15.2: what should we conclude?

One possible story is that siblings is an invalid instrument

Model

$\log(wage) = \beta_0 + \beta_1 educ + \gamma\, ability + v$. Instrument: *sibs*.



Less parental attention could link siblings to ability: $\text{Cov}(sibs, u) < 0$.

A possible upward distortion of the IV estimate

Two negative covariances can produce a positive IV distortion

Model

$\log(wage) = \beta_0 + \beta_1 educ + u.$

Assume $\text{Cov}(sibs, u) < 0$; maintain $\text{Cov}(sibs, educ) < 0$.

$$\text{plim}\,\hat{\beta}_{1,IV} = \beta_1 + \frac{\underbrace{\text{Cov}(sibs, u)}_{<0}}{\underbrace{\text{Cov}(sibs, educ)}_{<0}} > \beta_1$$

What the conditional sign story can establish

The observed ordering cannot diagnose either estimator's distortion

Conditional conclusion

$\log(wage) = \beta_0 + \beta_1 educ + u$. If both covariances are negative, $\text{plim}\,\hat{\beta}_{1,IV} > \beta_1$.

This hypothesis would push IV upward.

It does not establish downward OLS bias. The coefficient ordering alone cannot tell us which estimator is closer to the true return.

Contents

Intro 15-1a: binary instruments

1. Motivation: Omitted Variables in a Simple Regression Model
2. **Statistical Inference with the IV Estimator**
   - Precision and the return to education
   - **Binary instruments and binary treatments**
3. Properties of IV with a Poor Instrumental Variable
4. Computing R-Squared after IV Estimation
5. IV Estimation of the Multiple Regression Model

An instrument can also be binary

The same two conditions apply to numerical variables and indicators

Same requirements

$\text{Cov}(z, u) = 0$, $\text{Cov}(z, x) \neq 0$.

So far, schooling and the instruments are quantitative variables: years of father's schooling or number of siblings.

We now allow x, z, or both to be binary variables.

Binary instrument candidate z

Was the man born in the first quarter? It may shift years of schooling.

Binary regressor x

Did the man serve in the military? A draft lottery may shift that probability.

Quarter of birth and school leaving

Angrist and Krueger (1991): two clocks can change required schooling

How could your birth month affect how many years you spend in school?

Wage equation

$\log(wage) = \beta_0 + \beta_1 educ + u$; $frstqrt = 1$ for a January–March birth, zero otherwise.

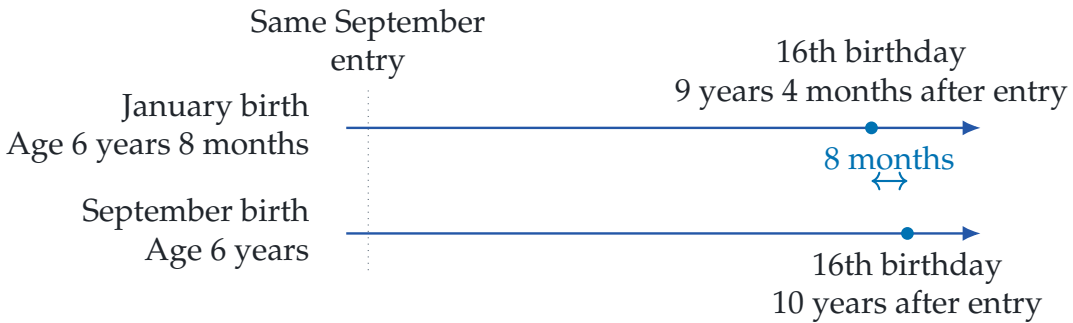
- **School entry** Historical US entry rules placed children into annual cohorts. Early-year births tended to begin at an older age.
- **School leaving** Attendance laws generally required staying until a specified birthday, often age 16; permission to leave did not require high-school completion.

School-grade progress and chronological age are different clocks.

The same school cohort, different birthdays

A January birthday reaches age sixteen eight months sooner

For this example, suppose children who turn six in the same calendar year enter together in September.



Earlier permission does not force dropout

Calendar time since entry is different from completed grades

Our common-entry example

Both enter in September of the year they turn six; legal leaving age is 16.

Birthday	Age at entry	Time until age 16
January	6 years 8 months	9 years 4 months
September	6 years	10 years

The January-born student has an earlier **opportunity** to leave. It does not force dropout or guarantee one fewer completed grade.

Who responds to permission to leave?

The mechanism concerns students constrained by the leaving-age rule

Two clocks
Same school-entry cohort; different dates at which age 16 is reached.

- **Would leave before the legal age** An earlier sixteenth birthday can allow less schooling.
- **Would continue through graduation** Earlier permission to leave need not change schooling.

Birth timing changes required schooling for potential dropouts.

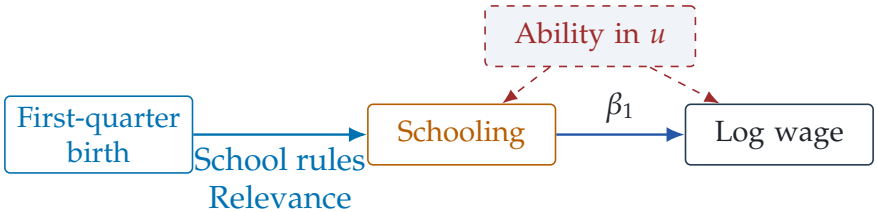
Because only part of the cohort responds, the average schooling difference can be small.

Birth quarter as a candidate instrument

Relevance links birth timing to schooling through the school rules

Model and relevance

$\log(wage) = \beta_0 + \beta_1 educ + u$; $\text{Cov}(frstqrt, educ) \neq 0$.



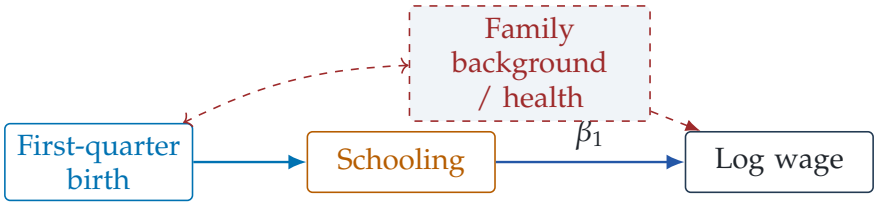
Birth quarter predicts only a small schooling difference: the first-stage R^2 is small. That can make the IV estimate imprecise.

Quarter of birth: what is still assumed?

The school-law mechanism does not settle other wage paths

Maintained exogeneity

$\log(wage) = \beta_0 + \beta_1 educ + u$; $\mathrm{Cov}(frstqrt, u) = 0$.



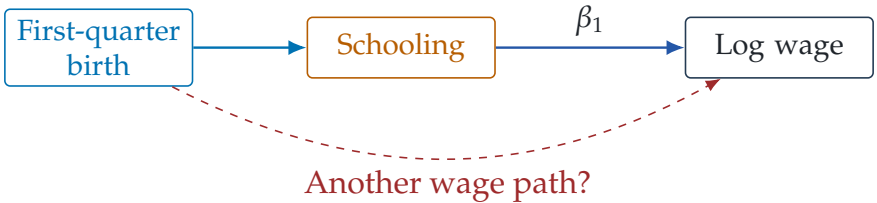
Seasonal family or health differences could threaten exogeneity.

Quarter of birth: another wage pathway?

Relative-age effects beyond completed schooling could threaten exclusion

Same model

$\log(wage) = \beta_0 + \beta_1 educ + u$. Instrument: first-quarter birth.



Relative-age effects on later wages beyond completed schooling could violate exclusion. This is a validity question, not an established path.

Quarter of birth: estimates and uncertainty

A binary instrument can identify a return but leave substantial uncertainty

Sample
247,199 men born in 1920–1929. $\log(wage) = \beta_0 + \beta_1 educ + u$.

Method	Schooling slope	SE	Approx. t
OLS	.0801	.0004	200
IV: first-quarter birth	.0715	.0219	3.26

$\hat{\beta}_{1,IV} = .0715 < .0801 = \hat{\beta}_{1,OLS}$: IV estimates about a 7.2% return.

The small first-stage relationship leaves IV much less precise than OLS, despite the very large sample.

Optional paper calculation

Binary regressor example: Veteran status and lifetime earnings

A binary treatment can be endogenous because people select into it

$$\log(earns) = \beta_0 + \beta_1 veteran + u \tag{15.18}$$

veteran = 1 for a Vietnam War veteran, zero otherwise. We want the effect of military service on earnings.

People who expect to gain more from military service may choose to join. Their other characteristics may also affect earnings.

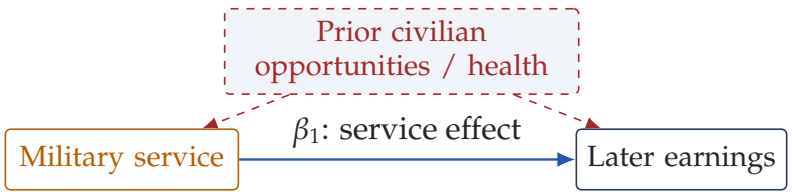
OLS can therefore mix the service effect with self-selection.

Self-selection can confound military service and earnings

Civilian opportunities can affect both the service decision and later earnings

Target

$\log(earns) = \beta_0 + \beta_1 veteran + u.$



We need variation unrelated to prior opportunities

Veterans and nonveterans can differ before service

Selection concern

$\log(earns) = \beta_0 + \beta_1 veteran + u$; $\text{Cov}(veteran, u) \neq 0$.

Veterans and nonveterans may differ in earning potential before service. OLS can mix the service effect with those prior differences.

What could change military service independently of prior opportunities?

The draft lottery as a natural experiment IV

Random draft priority can change the probability of service

Target
$\log(earns) = \beta_0 + \beta_1 veteran + u$. Instrument: draft lottery number.

- **Random assignment** In the randomized lotteries, numbers assigned draft priority independently of pre-existing earning potential.
- **Relevance** Low numbers raised the chance of being called to serve, so the number predicted veteran status.

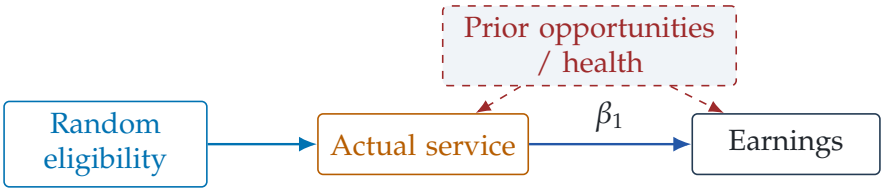
Let $z = eligible$ indicate a number low enough for call-up. **Eligibility** is the assignment; *veteran* is actual service.

Random eligibility, nonrandom service

Assignment can be random even when the response to it is not

Model

$\log(earns) = \beta_0 + \beta_1 veteran + u$; $z = eligible$ from randomized draft priority.



Some eligible men avoid or are exempted from service; some ineligible men volunteer. Actual service need not be random.

A natural experiment still needs exclusion

Draft avoidance could create another path to earnings

Target

$\log(earns) = \beta_0 + \beta_1 veteran + u$. Instrument: draft eligibility.

Suppose low draft numbers induce more schooling to avoid service.



Random assignment leaves an exclusion question

Behavior in response to assignment can create another earnings path

Target

$\log(earns) = \beta_0 + \beta_1 veteran + u$. Instrument: draft eligibility.

- The lottery could change earnings through schooling, as well as military service. That would violate exclusion in this model.
- Random assignment addresses prior differences; it does not eliminate additional paths created in response to assignment.

Contents

Intro 15-1b

1. Motivation: Omitted Variables in a Simple Regression Model
2. Statistical Inference with the IV Estimator
3. **Properties of IV with a Poor Instrumental Variable**
4. Computing R-Squared after IV Estimation
5. IV Estimation of the Multiple Regression Model

OLS and IV use different variation

Consistency requires the estimating variation to be uncorrelated with the structural error

$y = \beta_0 + \beta_1 x + u$, with $\text{Cov}(z, x) \neq 0$.

Which variable must be unrelated to the structural error?

OLS uses x $\quad \text{Cov}(x, u) = 0$.

IV uses z $\quad \text{Cov}(z, u) = 0$.

IV can be consistent even when $\text{Cov}(x, u) \neq 0$; its instrument must provide variation unrelated to u.

OLS and IV probability limits

Any remaining IV error correlation is divided by instrument relevance

$$y = \beta_0 + \beta_1 x + u.$$

$$\text{plim}\,\hat{\beta}_{1,OLS} = \beta_1 + \text{Corr}(x, u)\frac{\sigma_u}{\sigma_x} \tag{15.20}$$

$$\text{plim}\,\hat{\beta}_{1,IV} = \beta_1 + \frac{\text{Corr}(z, u)}{\text{Corr}(z, x)}\frac{\sigma_u}{\sigma_x} \tag{15.19}$$

IV divides the exogeneity violation by relevance; both correlations matter.

OLS limit | IV limit

Weak relevance can magnify bias

A small instrument-error correlation can become a large asymptotic bias

$$\text{Asymptotic bias}_{IV} = \frac{\text{Corr}(z, u)}{\text{Corr}(z, x)} \frac{\sigma_u}{\sigma_x}$$

Hold $\text{Corr}(z, u) = .02$ and $\sigma_u / \sigma_x = 1$ fixed.

What happens to the bias when instrument relevance falls?

A weaker denominator gives a larger bias

Hold the instrument-error correlation fixed

Asymptotic bias$_{IV}$ = .02/Corr(z, x), with $\sigma_u/\sigma_x = 1$.

Relevance Corr(z, x)	.50	.05
Asymptotic bias of IV	.04	.40

Weak relevance magnifies even a small exogeneity violation.

A weaker denominator gives a larger bias

Hold the instrument-error correlation fixed

Asymptotic bias$_{IV}$ = .02/Corr(z, x), with $\sigma_u/\sigma_x = 1$.

Relevance Corr(z, x)	.50	.05
Asymptotic bias of IV	.04	.40

Weak relevance magnifies even a small exogeneity violation.

Exact exogeneity gives zero asymptotic bias. Weak relevance can still make inference unreliable.

Example 15.3: Weak IV

Cigarette price barely predicts packs smoked

$\log(bwght) = \beta_0 + \beta_1 packs + u$. $n = 1{,}388$.

Higher cigarette prices might reduce smoking.

First stage $packs = \pi_0 + \pi_1 cigprice + r$.

First-stage term	Coefficient	SE
Cigarette price	.0003	.0008

Does the first stage show a clear price–smoking relationship?

The slope is close to zero.

Example 15.3: Weak IV

Cigarette price barely predicts packs smoked

$\log(bwght) = \beta_0 + \beta_1 packs + u$. $n = 1{,}388$.

Higher cigarette prices might reduce smoking.

First stage $packs = \pi_0 + \pi_1 cigprice + r$.

First-stage term	Coefficient	SE
Cigarette price	.0003	.0008

Does the first stage show a clear price–smoking relationship?

The slope is close to zero.

Its SE is much larger: there is little evidence that this price measure shifts smoking.

Example 15.3: an unstable IV estimate

Weak first-stage evidence cannot establish that smoking has no effect

$$\log(bwght) = \beta_0 + \beta_1 packs + u.$$

IV coefficient on packs	2.99
SE	8.70

Does this show that smoking has no effect?

Revisit the first stage | IV R-squared

Example 15.3: an unstable IV estimate

Weak first-stage evidence cannot establish that smoking has no effect

$$\log(bwght) = \beta_0 + \beta_1 packs + u.$$

IV coefficient on packs	2.99
SE	8.70

Does this show that smoking has no effect?

Uninformative and noisy estimate!

Weak first-stage evidence gives an uninformative IV design; a lack of significance does not establish a zero effect.

Revisit the first stage | IV R-squared

Contents

Intro 15-1c

1. Motivation: Omitted Variables in a Simple Regression Model
2. Statistical Inference with the IV Estimator
3. Properties of IV with a Poor Instrumental Variable
4. **Computing R-Squared after IV Estimation**
5. IV Estimation of the Multiple Regression Model

Structural IV fit: first estimate the coefficients

Use the instrument moment conditions

$y_i = \beta_0 + \beta_1 x_i + u_i.$

$$\hat{\beta}_{1,IV} = \frac{\widehat{\text{Cov}}(z, y)}{\widehat{\text{Cov}}(z, x)}, \qquad \hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV}\bar{x}.$$

These coefficients use the instrument. They do not minimize the squared errors from regressing y on observed x.

Structural IV fit: calculate residuals

Use actual x in the original structural equation

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

$$\hat{u}_i = y_i - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV} x_i.$$

Should this residual use actual x_i or a first-stage prediction?

Use actual x_i. The instrument enters through the IV coefficient estimates.

Even when a first stage computes those coefficients, use observed x_i in the structural residual.

Structural IV fit: compare with the mean

A negative R-squared means worse squared-error fit than predicting the mean

$y_i = \beta_0 + \beta_1 x_i + u_i.$

$$R^2_{IV} = 1 - \frac{\sum_i \hat{u}_i^2}{\sum_i (y_i - \bar{y})^2}.$$

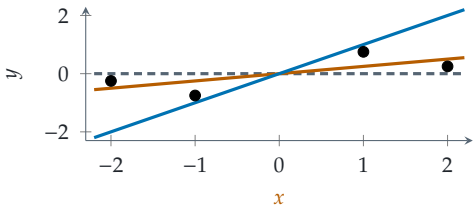
The denominator is the squared error from predicting $\bar{y}$ for everyone. OLS minimizes squared errors, so its $SSR \leq SST$. IV uses instrument moments, so $SSR_{IV} > SST$ is possible.

Negative fit and cross term | Population variance intuition

The IV line can fit worse than the mean

Structural R-squared measures sample fit; it does not establish instrument validity

$y_i = \beta_0 + \beta_1 x_i + u_i$. Suppose the structural slope is 1.



Prediction	Squared errors
Mean $\bar{y} = 0$	1.25
OLS: $.25x$	.625
IV: $1x$	6.25

$$R^2_{IV} = 1 - \frac{6.25}{1.25} = -4.$$

IV: $\sum z_i y_i / \sum z_i x_i = 10/10 = 1.$

Contents

Intro 15-2

1. Motivation: Omitted Variables in a Simple Regression Model
2. Statistical Inference with the IV Estimator
3. Properties of IV with a Poor Instrumental Variable
4. Computing R-Squared after IV Estimation
5. **IV Estimation of the Multiple Regression Model**
   - **Moment equations and conditional relevance**
   - College proximity and program eligibility

IV in a multiple regression model

An included exogenous regressor stays in the structural equation

$$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + u_1 \tag{15.22}$$

- **Endogenous regressor** y_2 may be correlated with u_1.
- **Included exogenous regressor** z_1 has its own effect and is uncorrelated with u_1.
- **Excluded instrument** z_2 is uncorrelated with u_1, excluded from this equation, and predicts y_2 after z_1.

The wage equation with an experience control

Education can remain endogenous after experience is included

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u_1 \tag{15.23}$$

- **Experience** Assume it is exogenous. It has its own wage effect and instruments itself.
- **Father's education** A further excluded instrument is needed for schooling. It must predict schooling after experience is held fixed.

Instrument roles | Partial relevance

Three population moment conditions

Assumptions supply equations for three structural coefficients

Wage mapping

$y_1 = \log(wage)$, $y_2 = educ$, $z_1 = exper$, $z_2 = fatheduc$.

$$E(u_1) = 0, \quad \mathrm{Cov}(z_1, u_1) = 0, \quad \mathrm{Cov}(z_2, u_1) = 0 \tag{15.24}$$

$$\begin{aligned} E(y_1 - \beta_0 - \beta_1 y_2 - \beta_2 z_1) &= 0, \\ E[z_1(y_1 - \beta_0 - \beta_1 y_2 - \beta_2 z_1)] &= 0, \\ E[z_2(y_1 - \beta_0 - \beta_1 y_2 - \beta_2 z_1)] &= 0. \end{aligned}$$

These equations identify $\beta_0, \beta_1, \beta_2$ when the relevance/rank condition makes the system nonsingular.

The sample method of moments

Solving these equations gives the multiple-regression IV estimator

$$\hat{u}_{i1} = y_{i1} - \hat{\beta}_0 - \hat{\beta}_1 y_{i2} - \hat{\beta}_2 z_{i1}.$$

$$\begin{aligned} \sum_{i=1}^{n} \hat{u}_{i1} &= 0 \quad \text{mean error} \\ \sum_{i=1}^{n} z_{i1}\hat{u}_{i1} &= 0 \quad \text{experience} \\ \sum_{i=1}^{n} z_{i2}\hat{u}_{i1} &= 0 \quad \text{father's schooling} \end{aligned} \tag{15.25}$$

Solve these three sample equations for the three IV coefficients.

Relevance after including a control

The instrument must predict schooling beyond experience

$$y_2 = \pi_0 + \pi_1 z_1 + \pi_2 z_2 + v_2 \tag{15.26}$$

$y_2 = educ$, $z_1 = exper$, $z_2 = fatheduc$.

$$\pi_2 \neq 0 \tag{15.27}$$

Estimate this first stage and test the father's-schooling coefficient. The required association remains *after experience is held fixed*.

The first stage with several controls

The excluded instrument has a partial effect beyond all included controls

Structural model

$y_1 = \beta_0 + \beta_1 y_2 + \beta_2 z_1 + \cdots + \beta_k z_{k-1} + u_1.$

$$y_2 = \pi_0 + \pi_1 z_1 + \cdots + \pi_{k-1} z_{k-1} + \pi_k z_k + v_2 \tag{15.30}$$

In the college example, $y_2 = educ$, and the excluded instrument is $z_k = nearc4$.

$\pi_k \neq 0$ means the instrument predicts schooling after the included controls.

Contents

Intro 15-2: Example 15.4

1. Motivation: Omitted Variables in a Simple Regression Model
2. Statistical Inference with the IV Estimator
3. Properties of IV with a Poor Instrumental Variable
4. Computing R-Squared after IV Estimation
5. **IV Estimation of the Multiple Regression Model**
   - Moment equations and conditional relevance
   - **College proximity and program eligibility**

Example 15.4: college proximity

Access may shift schooling, but location needs an exclusion argument

$$\log(wage) = \beta_0 + \beta_1 educ + \text{control effects} + u.$$

- **Sample** 3,010 men; wages and education measured in 1976; the same observations and controls in OLS and IV.
- **Instrument** `nearc4`: lived near a four-year college in 1966.
- **Controls**
  - Experience and its square: `exper`, `exper`^{2}; black and South indicators: `black`, `south`.
  - Standard metropolitan statistical area (SMSA): current `smsa` and 1966 `smsa66`; 1966 regions: `reg662`, ..., `reg669`.

Example 15.4: first-stage evidence

Nearby colleges predict more schooling after controls

$$educ = \pi_0 + \pi_{near}\, nearc4 + \text{control effects} + r.$$

Excluded instrument	Coefficient	SE	t for zero
Near a college	.320	.088	3.64

Same 3,010 men and controls as the outcome equation.

Nearby colleges predict about one third of an additional schooling year, holding the controls fixed.

Example 15.4: first-stage evidence

Nearby colleges predict more schooling after controls

$$educ = \pi_0 + \pi_{near}\, nearc4 + \text{control effects} + r.$$

Excluded instrument	Coefficient	SE	t for zero
Near a college	.320	.088	3.64

Same 3,010 men and controls as the outcome equation.

Nearby colleges predict about one third of an additional schooling year, holding the controls fixed.

This supports conditional relevance. $\pi_{near} \neq 0$ in the population first stage after the controls.

Example 15.4: OLS and IV

Read the coefficient and its SE together

Target

$\log(wage) = \beta_0 + \beta_1 educ + \text{controls} + u$.

Table 15.1	OLS	IV (*nearc4*)
Schooling coefficient	.075	.132
SE	.003	.055

Example 15.4: OLS and IV

Read the coefficient and its SE together

Target
$\log(wage) = \beta_0 + \beta_1 educ + \text{controls} + u$.

Table 15.1	OLS	IV (*nearc4*)
Schooling coefficient	.075	.132
SE	.003	.055

IV gives a larger point estimate with much greater uncertainty.

The instrument can predict the outcome too

A first-stage coefficient and an outcome reduced-form coefficient have different units

Structural model

$y_1 = \beta_0 + \beta_1 y_2 + \text{controls} + u_1$.

First stage

How does z_k change y_2, after the controls?

Outcome reduced form

How does the same z_k change y_1, after the same controls?

Use a program offer to distinguish *participation* from the *outcome* response.

Program eligibility and participation

Distinguish eligibility, take-up and the outcome

$z_k = 1$: eligible; $y_2 = 1$: participates; y_1: outcome. Some eligible people do not participate.

$$
\begin{aligned}
y_1 &= \beta_0 + \beta_1 y_2 + \text{controls} + u_1 && \text{structural} \\
y_2 &= \pi_0 + \pi_k z_k + \text{controls} + v_2 && \text{first stage} \\
y_1 &= \gamma_0 + \gamma_k z_k + \text{controls} + e_1 && \text{outcome reduced form}
\end{aligned}
$$

The three coefficients describe different responses: β_1, outcome effect of participation; π_k, participation after eligibility; γ_k, outcome after eligibility.

The outcome also has a reduced form

Under IV exogeneity and exclusion, eligibility works through participation

$$
\begin{aligned}
y_1 &= \beta_0 + \beta_1 y_2 + \text{controls} + u_1, \\
y_2 &= \pi_0 + \pi_k z_k + \text{controls} + v_2, \\
y_1 &= \gamma_0 + \gamma_k z_k + \text{controls} + e_1. \qquad (15.32)
\end{aligned}
$$



$\gamma_k = \beta_1 \pi_k$: combined outcome response to eligibility

Reduced-form substitution | Exclusion and exogeneity

Offering a program and participating in it

An eligibility effect allows for nonparticipation

Same model

$y_1 = \beta_0 + \beta_1 y_2 + \text{controls} + u_1$, with $\gamma_k = \beta_1 \pi_k$.

- π_k: eligibility's change in participation probability, after controls.
- β_1: participation's effect on the outcome under the constant-effect model.
- γ_k: eligibility's effect on the outcome, allowing for nonparticipation: the *intention-to-treat* effect.

Suppose $\pi_k = .20$ and $\beta_1 = 5$: $\gamma_k = 5 \times .20 = 1$ outcome unit.

A practical guide to a good instrument

Assess the design and the evidence together

- **Start with the population/structural model** State the causal coefficient and what remains in u.
- **Mechanism** Explain how z changes the endogenous regressor.
- **First stage** Check the sign, magnitude and strength after the same controls.
- **Exogeneity** Defend both the absence of other outcome paths and the absence of shared omitted causes.
- **Uncertainty** Report appropriate IV uncertainty; a strong first stage alone does not establish validity.

Weak instruments

What you should be able to do

Prepare to assess and interpret an instrumental-variables argument

- State the outcome equation, target effect, endogenous variable and be able to propose reasonable IV candidate in different context.
- Explain exogeneity/exclusion and relevance in the application, including relevance after controls.
- Interpret an IV estimate, its uncertainty and evidence of a weak instrument.

Deriving IV consistency is not required for assessment; the argument remains available for further reading.

Where to read further

Match today's questions to further reading

Topic	Intro 6e
Assumptions and estimation	15-1, pp.462–465
Uncertainty and examples	15-1a, pp.466–469
Poor instruments	15-1b, pp.469–471
Interpreting IV R-squared	15-1c, p.471
IV with controls	15-2, pp.471–475

Optional formal detail: the graduate book's Chapter 5 and the IV supplement.

Optional explanations

Next lecture

More than one excluded instrument leads to two-stage least squares

- **Lecture 4b** Two-stage least squares, weak instruments, measurement error, specification tests and robust inference.
- Read *Introductory Econometrics* 15-3–15-6, including endogeneity and overidentification tests.
- Problem Set 2 due Friday 23 October 2026; Midterm: Monday 26 October 2026, 09:00–11:00. Coverage: L1–L4.

Adapted from slides by Shuyang Sheng and Qihui Chen

Optional explanations

Short derivations for the main lecture

- Proxy substitution
- Structural residuals
- R-squared and correlation
- OLS probability limit
- IV probability limit
- Simple-IV consistency
- Negative IV R-squared
- Sample cross term
- Variance decomposition
- First-stage slope and relevance
- Optional paper calculation

Extended proofs and applications: *Instrumental Variables Supplement*, contents by Chapter 15 topic.

The error second moment

A conditional restriction implies an unconditional second moment

$$E(u^2 \mid z) = \sigma^2 = \text{Var}(u). \tag{15.11}$$

The conditional-second-moment restriction $E(u^2 \mid z) = \sigma^2$ is assumed. Also assume $E(u) = 0$.

$$E(u^2) = E\Big[E(u^2 \mid z)\Big] = E(\sigma^2) = \sigma^2.$$

From second moment to variance | Back to IV inference assumptions

From the second moment to the variance

Mean-zero error makes its second moment equal its variance

$E(u^2) = \sigma^2$ and $E(u) = 0$.

$$\text{Var}(u) = E(u^2) - [E(u)]^2 = \sigma^2 - 0 = \sigma^2.$$

Thus $E(u^2 \mid z) = \sigma^2 = \text{Var}(u)$: the second equality follows from LIE and the mean-zero error.

$\text{Cov}(z, u) = 0$ alone does not imply $E(u \mid z) = 0$, or a constant conditional second moment.

Back to IV inference assumptions

What the proxy changes in the equation

Proxy review: the remaining error must be exogenous

$$
\begin{aligned}
y &= \beta_0 + \beta_1 educ + \gamma \underbrace{(\delta_0 + \delta_p p + r)}_{q} + v \\
&= (\beta_0 + \gamma\delta_0) + \beta_1 educ + \gamma\delta_p p + \underbrace{(\gamma r + v)}_{e}.
\end{aligned}
$$

Under the proxy conditions, $E(e) = 0$ and $\text{Cov}(educ, e) = \text{Cov}(p, e) = 0$. With full rank and the usual sampling and moment conditions, OLS on schooling and the proxy consistently estimates β_1.

Back to proxy conditions

Structural residuals estimate structural noise

Use the observed regressor when calculating the residual

$$\hat{u}_i = y_i - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV} x_i$$

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^{n} \hat{u}_i^2$$

- Two fitted structural coefficients: intercept and slope.
- z enters through the coefficient estimates. The adjustment does not make $\hat{\sigma}^2$ exactly unbiased.

Back to estimated IV variance

R-squared equals squared correlation

Simple regression with an intercept; either direction gives the same R-squared

Centered sums $S_{ab} = \sum_i (a_i - \bar{a})(b_i - \bar{b})$.

$$\hat{\pi}_1 = \frac{S_{xz}}{S_{zz}}, \qquad R^2_{x \text{ on } z} = \frac{\hat{\pi}_1^2 S_{zz}}{S_{xx}}$$

$$R^2_{x \text{ on } z} = \frac{S_{xz}^2}{S_{xx} S_{zz}} = \widehat{\text{Corr}}(x, z)^2 = R^2_{z \text{ on } x}.$$

Estimates ρ_{xz}^2, not ρ_{xz}. Requires an intercept and variation in both variables; does not generally hold with multiple regressors.

Back to the IV standard error

Derivation of the OLS probability limit

Expand the covariance; then use the correlation definition

For $y = \beta_0 + \beta_1 x + u$, use random sampling, finite moments and $\text{Var}(x) > 0$.

$$\begin{aligned} \text{plim}\, \hat{\beta}_{1,OLS} &= \frac{\text{Cov}(x, y)}{\text{Var}(x)} \\ &= \beta_1 + \frac{\text{Cov}(x, u)}{\sigma_x^2} \\ &= \beta_1 + \text{Corr}(x, u)\frac{\sigma_u}{\sigma_x}. \qquad (15.20) \end{aligned}$$

Back to the OLS probability limit

Derivation of the IV probability limit

Relevance permits division; exogeneity removes the distortion

For $y = \beta_0 + \beta_1 x + u$, use random sampling, finite moments and $\text{Cov}(z, x) \neq 0$.

$$\begin{aligned} \text{plim}\, \hat{\beta}_{1,IV} &= \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)} = \beta_1 + \frac{\text{Cov}(z, u)}{\text{Cov}(z, x)} \\ &= \beta_1 + \frac{\text{Corr}(z, u)\sigma_z\sigma_u}{\text{Corr}(z, x)\sigma_z\sigma_x} \\ &= \beta_1 + \frac{\text{Corr}(z, u)}{\text{Corr}(z, x)}\frac{\sigma_u}{\sigma_x}. \qquad (15.19) \end{aligned}$$

Back to the IV probability limit

Consistency of simple IV

Scalar case of Intro Theorem 15A.1

$$y = \beta_0 + \beta_1 x + u, \quad E(u) = 0, \quad \mathrm{Cov}(z, u) = 0, \quad \mathrm{Cov}(z, x) \neq 0$$

With random sampling and the required finite moments,

$$\hat{\beta}_{1,IV} \xrightarrow{p} \frac{\mathrm{Cov}(z, y)}{\mathrm{Cov}(z, x)} = \beta_1 + \frac{\mathrm{Cov}(z, u)}{\mathrm{Cov}(z, x)} = \beta_1.$$

Thus $\hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV}\bar{x} \xrightarrow{p} \beta_0$.
Intro Theorem 15A.1 gives the general consistency result under Assumptions 2SLS.1–2SLS.4.

Back to IV consistency

Why IV R-squared can be negative

The IV line can fit worse than the mean-only prediction

$$\hat{u}_i^{IV} = y_i - \hat{\beta}_0^{IV} - \hat{\beta}_1^{IV} x_i, \qquad R^2 = 1 - \frac{SSR_{IV}}{SST}$$

Mean-only benchmark Predicting $\bar{y}$ for everyone gives squared error SST.

OLS With an intercept, least squares guarantees $SSR_{OLS} \leq SST$.

IV Instrument moment conditions do not minimize squared errors. Structural SSR_{IV} can exceed SST, giving negative R^2.

Sample cross term | Back to IV R-squared

Why IV R-squared can be negative

The OLS decomposition contains a cross term for IV

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{u}_i, \qquad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$$SST = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{u}_i^2 + 2\sum_i (\hat{y}_i - \bar{y})\hat{u}_i$$

- OLS residuals are orthogonal to x in the sample, even if x is endogenous; the cross term vanishes.
- Simple IV makes $\sum_i z_i \hat{u}_i = 0$; it does not generally make $\sum_i x_i \hat{u}_i = 0$.

These residuals use actual x_i, not first-stage fitted $\hat{x}_i$. Thus structural SSR can exceed SST, giving $1 - SSR/SST < 0$.

Back to IV R-squared

Why structural error variance can exceed outcome variance

The structural component and error can offset each other

$$y = \beta_0 + \beta_1 x + u.$$

$$\mathrm{Var}(y) = \beta_1^2 \mathrm{Var}(x) + \mathrm{Var}(u) + 2\beta_1 \mathrm{Cov}(x, u).$$

If $\mathrm{Cov}(x, u) = 0$, total variance sums two nonnegative pieces.

A sufficiently negative cross term can make total variance smaller than either component. Negative structural R^2 is possible, not inevitable.

Variance derivation | Numerical example | Back to structural IV fit

Deriving the variance decomposition

The cross term is present before exogeneity is imposed

Centered structural equation

$y - E(y) = \beta_1[x - E(x)] + [u - E(u)]$.

$$[y - E(y)]^2 = \beta_1^2[x - E(x)]^2 + [u - E(u)]^2 + 2\beta_1[x - E(x)][u - E(u)].$$

$$\operatorname{Var}(y) = \beta_1^2 \operatorname{Var}(x) + \operatorname{Var}(u) + 2\beta_1 \operatorname{Cov}(x, u).$$

Taking expectations yields the identity. Only the extra condition $\operatorname{Cov}(x, u) = 0$ removes its last term.

Back to the variance identity

A negative population fit measure

The components can offset each other

$$\text{Var}(y) = \beta_1^2 \text{Var}(x) + \text{Var}(u) + 2\beta_1 \text{Cov}(x, u).$$

Suppose $\beta_1 = 1$, $\text{Var}(x) = \text{Var}(u) = 1$, and $\text{Cov}(x, u) = -.75$.

$$\text{Var}(y) = 1 + 1 + 2(-.75) = .5, \qquad 1 - \frac{\text{Var}(u)}{\text{Var}(y)} = -1.$$

Under consistent IV and finite moments, this ratio is the large-sample limit of structural R^2, rather than separate variance shares.

Back to variance intuition

Binary-instrument IV

Compare outcome changes with schooling changes

$$\hat{\beta}_1 = \frac{\bar{y}_1 - \bar{y}_0}{\bar{x}_1 - \bar{x}_0}$$

- This identity applies to a single binary instrument and an intercept, with no other controls.
- A hypothetical outcome gap of 0.08 and schooling gap of 1 gives 0.08 log points per year.
- Interpretation as a structural effect still requires the maintained instrument assumptions.

Back to the Wald estimate

Why the first-stage slope measures relevance

Covariance with the instrument isolates its slope

Population first stage

$x = \pi_0 + \pi_1 z + v$, with $\mathrm{Cov}(z, v) = 0$ by linear projection.

$$\mathrm{Cov}(z, x) = \mathrm{Cov}(z, \pi_0 + \pi_1 z + v) = \pi_1 \, \mathrm{Var}(z).$$

$$\pi_1 = \frac{\mathrm{Cov}(z, x)}{\mathrm{Var}(z)}, \qquad \pi_1 \neq 0 \iff \mathrm{Cov}(z, x) \neq 0.$$

Requires $\mathrm{Var}(z) > 0$ (implied by relevance: a constant z has zero covariance with x). For MROZ, z = *fatheduc*, x = *educ*.

Back to MROZ first-stage evidence

Two responses to the instrument

The outcome reduced form is the response of wages to the instrument

Same structural equation

$\log(wage) = \beta_0 + \beta_1 educ + u$.

First stage

Compare schooling across instrument groups: $\Delta\overline{educ}$.

Outcome reduced form

Compare log wages across the same groups: $\Delta\overline{\log(wage)}$.

Under the IV assumptions, their ratio gives log-wage units per schooling year.

Back to birth-quarter results | Paper sample calculation

Birth quarter: scale two differences

The Wald estimate uses a small first-stage schooling difference

$\hat{\beta}_{IV} = \Delta\overline{\log(wage)} / \Delta\overline{educ}$. Men born 1930–1939.

Published Table III, Panel B	Estimate
Q1-minus-other log-wage difference	−.01110
Q1-minus-other schooling difference	−.1088

$$\hat{\beta}_{IV} = \frac{-.01110}{-.1088} \approx .1020$$

Angrist and Krueger (1991), Table III, Panel B.

Binary-instrument covariance identity

Back to the two responses

The probability limit of OLS

Correlation with the structural error changes the limit

$$y = \beta_0 + \beta_1 x + u$$

$$\hat{\beta}_{1,OLS} = \frac{\widehat{\text{Cov}}(x, y)}{\widehat{\text{Var}}(x)}$$

$$\text{plim}\, \hat{\beta}_{1,OLS} = \beta_1 + \text{Corr}(x, u)\frac{\sigma_u}{\sigma_x} \tag{15.20}$$

OLS relates y to x. The error covariance is therefore $\text{Cov}(x, u)$.

Derivation of 15.20 | Back to OLS and IV comparison

The probability limit of IV

Instrument-error correlation is divided by instrument relevance

$$y = \beta_0 + \beta_1 x + u$$

$$\hat{\beta}_{1,IV} = \frac{\widehat{\text{Cov}}(z, y)}{\widehat{\text{Cov}}(z, x)}$$

$$\text{plim}\, \hat{\beta}_{1,IV} = \beta_1 + \frac{\text{Corr}(z, u)}{\text{Corr}(z, x)} \frac{\sigma_u}{\sigma_x} \tag{15.19}$$

IV scales $\text{Cov}(z, u)$ by instrument relevance, $\text{Cov}(z, x)$.

Derivation of 15.19 | Back to OLS and IV comparison

Reduced form: substitute the first stage

The excluded instrument enters through the endogenous regressor

$$y_1 = \beta_0 + \beta_1 y_2 + \sum_{j=1}^{k-1} \beta_{j+1} z_j + u_1,$$
$$y_2 = \pi_0 + \sum_{j=1}^{k-1} \pi_j z_j + \pi_k z_k + v_2.$$

$$y_1 = \beta_0 + \beta_1 \left(\pi_0 + \sum_{j=1}^{k-1} \pi_j z_j + \pi_k z_k + v_2 \right) + \sum_{j=1}^{k-1} \beta_{j+1} z_j + u_1.$$

Collect the coefficients | Back to the outcome reduced form

Reduced form: collect the coefficients

Only the excluded instrument has no separate structural effect

$$y_1 = \gamma_0 + \sum_{j=1}^{k-1} \gamma_j z_j + \gamma_k z_k + e_1.$$

$$\begin{aligned} \gamma_0 &= \beta_0 + \beta_1 \pi_0, \\ \gamma_j &= \beta_{j+1} + \beta_1 \pi_j \quad (j = 1, \ldots, k-1), \\ \gamma_k &= \beta_1 \pi_k, \qquad e_1 = u_1 + \beta_1 v_2. \end{aligned}$$

For the excluded instrument, $\gamma_k / \pi_k = \beta_1$ when $\pi_k \neq 0$.

Back to the outcome reduced form

Why exclusion matters for the ratio

A direct instrument effect adds a second term

$$y_1 = \beta_0 + \beta_1 y_2 + \delta z_k + \text{controls} + u_1.$$

With a direct structural δz_k effect, substitute the first stage:

$$\gamma_k = \delta + \beta_1 \pi_k, \qquad \frac{\gamma_k}{\pi_k} = \beta_1 + \frac{\delta}{\pi_k}.$$

The ratio isolates β_1 only when exclusion removes δ.

Why exogeneity matters | Back to the outcome reduced form

Why exogeneity matters for the reduced form

The new error must remain orthogonal to the exogenous regressors

$e_1 = u_1 + \beta_1 v_2$, from substituting the first stage.

$$\text{Cov}(z_j, e_1) = \underbrace{\text{Cov}(z_j, u_1)}_{\text{assumed } 0} + \beta_1 \underbrace{\text{Cov}(z_j, v_2)}_{\text{projection: } 0} = 0.$$

Structural-error orthogonality is assumed. First-stage residual orthogonality follows from projection.

With rank, sampling and moment conditions, OLS consistently estimates the outcome reduced form.

Back to the outcome reduced form

Example 15.1: controls are instruments

Instrument education while retaining the experience effects

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u.$$

Outcome-equation variable	Instrument role
1, *exper*, $exper^2$	Included: instrument themselves
educ	Endogenous variable to instrument
fatheduc	Excluded from the outcome equation

The excluded instrument must have no remaining direct wage effect and no association with u, after the model's controls.

Back to the wage model with controls

Example 15.1 revisited: partial relevance

A raw correlation is not the required first-stage evidence

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \theta\, fatheduc + r.$$

$$\theta \neq 0.$$

- This is the **population linear projection**; its residual is orthogonal to its regressors by construction.
- $\theta \neq 0$ is an additional relevance condition, not a consequence of projection.
- Test the father's schooling coefficient in the first stage with the same experience controls.

Back to the wage model with controls

When a proxy removes omitted-variable bias

The proxy must remove the ability variation related to schooling

Structural equation

$y = \log(wage) = \beta_0 + \beta_1 educ + \gamma q + v$, with $E(v \mid educ, q) = 0$.

1. Redundancy $E(y \mid educ, q, p) = E(y \mid educ, q)$.

Once ability and schooling are held fixed, the proxy adds no wage effect.

2. Sufficient capture

$$L(q \mid 1, educ, p) = L(q \mid 1, p) = \delta_0 + \delta_p p.$$

After the proxy predicts ability, schooling must not predict the ability left over. Merely being correlated with ability is insufficient.

Proxy substitution | Quiz 2: can we test proxy conditions? | Back to proxy and IV

Quiz 2: can we test proxy conditions?

Unobserved skill prevents a direct check using this dataset alone

$output = \beta_0 + \beta_1 AIuse + \gamma skill + v$. Experience is the proxy.

Does a significant experience coefficient establish the proxy conditions?

No. Unobserved true skill prevents us from directly checking its remaining relationship with AI use after experience, or holding skill fixed to test redundancy.

A significant proxy coefficient does not show that it captures the skill differences between users and nonusers.

Evidence for a proxy or another design | Back to proxy conditions

Quiz 2: what evidence would help?

Validate how much skill the proxy captures, or change the design

$$output = \beta_0 + \beta_1 AIuse + \gamma skill + v.$$

- **Reasoning** Explain why remaining skill should no longer be related to AI use after controlling for the proxy.
- **External validation** When available, use an independent skill measure to assess the proxy argument.
- **Different design** A randomized rollout supplies another route to the causal effect.

Back to proxy conditions

Why IV can be consistent

The sample ratio approaches the population ratio

$$\hat{\beta}_{1,IV} = \frac{\widehat{\text{Cov}}(z, y)}{\widehat{\text{Cov}}(z, x)} \xrightarrow{p} \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)} = \beta_1$$

- Random sampling and suitable finite moments give convergence of the sample covariances.
- Exogeneity gives the right target; nonzero relevance keeps the population denominator away from zero.
- Consistent does not mean unbiased in a finite sample.

Back to the two IV conditions | Consistency proof